Lecture Scribe CSE 400

Tutorial_2_Solution.pdf
PDF

Generate a lecture scribe for exam preparation based on provided pdf. Write the scribe by reason step by step through definations, notation, proofs and example solutions exactly given in the provided context. Give step-by-step derivations for problems and examples solution. Write scribe with a clear logical flow so that student can understand easily. Do not include any points or derivations which is not present in the context.

Here is your **Lecture Scribe for CSE 400 – Tutorial 2 (Exam Preparation)**
Prepared strictly from the provided solution document, with step-by-step reasoning, derivations, and logical flow exactly as given.

CSE 400: Fundamentals of Probability in Computing

Tutorial 2 – Lecture Scribe (Step-by-Step)

Q1. Random Point on a Line Segment

Problem Interpretation

A point is chosen uniformly at random on a line segment of length L.

Let:

- The point be at distance x from one end.
- Then $0 < x < L$.
- The two segments formed are:
  - x
  - $L - x$

The ratio of the shorter to the longer segment is:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We must find:

$$P(R < 1/4)$$

Logical Case Division

We divide based on midpoint $L/2$.

Case 1: $x \leq L/2$

Shorter segment = x

Longer segment = $L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

Multiply both sides:

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > L/2$

Shorter segment = $L - x$

Longer segment = x

$$\frac{L-x}{x} < \frac{1}{4}$$

Multiply:

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\left(\frac{L}{5} - 0\right) + \left(L - \frac{4L}{5}\right) = \frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since uniform over length L:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayes Classification with Normal Densities

Given

White region:

$$X|W \sim N(4, 4), \quad \sigma_1 = 2$$

Black region:

$$X|B \sim N(6, 9), \quad \sigma_2 = 3$$

Prior probabilities:

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observed:

$$X = 5$$

We want equal posterior error:

$$P(B|X = 5) = P(W|X = 5) = \frac{1}{2}$$

Bayes Formula

$$P(B|X = 5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1 - \alpha) f(5|W)}$$

Set equal to 1/2.

Normal Density Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute densities:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{\alpha e^{-1/18}}{3\sqrt{2\pi}} \Bigg/ \left(\frac{\alpha e^{-1/18}}{3\sqrt{2\pi}} + \frac{(1-\alpha)e^{-1/8}}{2\sqrt{2\pi}} \right) = \frac{1}{2}$$

Simplifies to:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solve

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Q3. Fire Station Location

(a) Uniform(0, A)

$$X \sim \text{Uniform}(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[|X - a|]$$

Split integral:

$$\int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$$

Differentiate:

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a - 2A)$$

Set zero:

$$4a = 2A$$

$$a = \frac{A}{2}$$

Station at midpoint.

(b) Exponential(λ)

$$f_X(x) = \lambda e^{-\lambda x}$$

Differentiate expectation:

$$\frac{d}{da}E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = 1/2$$

$$-\lambda a = \ln(1/2) = -\ln 2$$

$$a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

Two battery types:

$$P(T = i) = p_i$$

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

Similarly:

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Poisson

$$X \sim \text{Poisson}(\lambda_1)$$

$$Y \sim \text{Poisson}(\lambda_2)$$

Independent.

Find:

$$P(X = k | X + Y = n)$$

Step 1

$$P(X = k, Y = n - k) = P(X = k) P(Y = n - k)$$

Substitute Poisson PMF:

$$= \frac{e^{-\lambda_1} \lambda_1^k}{k!} \cdot \frac{e^{-\lambda_2} \lambda_2^{n-k}}{(n-k)!}$$

Denominator:

$$P(X + Y = n) = \frac{e^{-(\lambda_1 + \lambda_2)} (\lambda_1 + \lambda_2)^n}{n!}$$

Final Simplified Form

$$P(X = k|X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Binomial structure appears.

Q6. Transformation of Random Variable

$$X \sim \text{Uniform}[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

Piecewise function:

- x for $[-2, -1]$
- 0 for $(-1, 1)$
- x for $[1, 2]$

CDF

Compute piecewise:

- $y < -2$: 0
- $-2 \leq y < -1$: $\frac{y+2}{4}$
- $-1 \leq y < 0$: $1/4$
- At $y = 0$: jump of $1/2$

$$F_Y(0) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

- $0 < y < 1$: $3/4$
- $1 \leq y \leq 2$: $\frac{y+2}{4}$
- $y > 2$: 1

PDF

$$f_Y(y) = \frac{1}{4} \quad \text{for } (-2, -1) \text{ and } (1, 2)$$

Point mass:

$$P(Y = 0) = 1/2$$

Q7. Stock Model

$$S_{1000} = S_0 u^k d^{1000-k}$$

Require:

$$u^k d^{1000-k} \geq 1.30$$

Take logs:

$$k \ln u + (1000 - k) \ln d \geq \ln(1.30)$$

Solve:

$$k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d}$$

Substitute values:

$$k \geq 469.4$$

Thus:

$$k \geq 470$$

Since:

$$k \sim \text{Binomial}(1000, 0.52)$$

Probability:

$$\sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Distribution

$$X \sim \text{Exponential}(1/2)$$

PDF:

$$f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1} \approx 0.3679$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

Region:

$$x^2 + y^2 \leq z^2$$

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate using Leibniz rule:

$$\frac{d}{dz}\sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final PDF:

$$f_Z(z) = z \int_{-z}^{z} f_X(x) \frac{f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of M_n, M_{n+1}

$$M_n = \max(X_1, ..., X_n)$$

$$M_{n+1} = \max(M_n, X_{n+1})$$

CDF of M_n

$$P(M_n \leq x) = F(x)^n$$

Case 1: $x \leq y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(y)^{n+1}$$

Final Joint CDF

$$P(M_n \leq x, M_{n+1} \leq y) = \begin{cases} F(x)^n F(y), & x \leq y \\ F(y)^{n+1}, & x > y \end{cases}$$

End of Tutorial 2 Lecture Scribe

If you want, I can now:

- Convert this into **super compact revision notes (2–3 pages)**
- Or make a **formula sheet only**
- Or create **exam-type practice questions with solutions** for fast revision before exam**

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CSE 400: Fundamentals of Probability in Computing

Tutorial-2 Lecture Scribe (Step-by-Step Derivations)

Q1. Random Point on a Line Segment

Interpretation

A point is chosen uniformly at random on a line segment of length L.

Let:

- The point be at distance x from one end.
- $0 < x < L$.

This creates two segments:

- Length x
- Length $L - x$

The ratio of the shorter to the longer segment is:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We need:

$$P(R < 1/4)$$

Case Division (Based on Midpoint)

Case 1: $x \leq L/2$

Shorter segment = x

Longer segment = $L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

Multiply both sides:

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > L/2$

Shorter segment = $L - x$

Longer segment = x

$$\frac{L-x}{x} < \frac{1}{4}$$

Multiply:

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\left(\frac{L}{5} - 0\right) + \left(L - \frac{4L}{5}\right) = \frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since uniform over total length L:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Equal Posterior Error (Bayes Theorem)

Given

White region:

$$X|W \sim N(4,4), \quad \sigma_1 = 2$$

Black region:

$$X|B \sim N(6,9), \quad \sigma_2 = 3$$

Prior probabilities:

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observed:

$$X = 5$$

Condition for equal error:

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Bayes Formula

$$P(B|X=5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)}$$

Set equal to $1/2$.

Normal Density Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute densities:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solve

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Q3. Optimal Location Problem

(a) Uniform(0,A)

$$X \sim Uniform(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[|X - a|]$$

Break integral:

$$\int_0^a (a-x)\frac{1}{A}dx + \int_a^A (x-a)\frac{1}{A}dx$$

Differentiate:

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a - 2A)$$

Set equal to zero:

$$4a = 2A$$

$$a = \frac{A}{2}$$

(b) Exponential(λ)

$$f_X(x) = \lambda e^{-\lambda x}$$

Derivative:

$$\frac{d}{da}E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = 1/2$$

$$a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

Let lifetime X.

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Poisson

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

Independent.

Find:

$$P(X = k|X + Y = n)$$

Using:

$$P(X = k, Y = n - k) = P(X = k)P(Y = n - k)$$

Substitute Poisson PMFs and simplify:

$$= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Q6. Transformation of Random Variable

$$X \sim Uniform[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

Piecewise $g(x)$:

- x for $[-2, -1]$
- 0 for $(-1, 1)$
- x for $[1, 2]$

CDF $F_Y(y)$

- $y < -2$: 0
- $-2 \leq y < -1$: $\frac{y+2}{4}$
- $-1 \leq y < 0$: $1/4$
- At $y = 0$:

$$P(Y = 0) = \frac{2}{4} = \frac{1}{2}$$

$$F_Y(0) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

- $0 < y < 1$: $3/4$
- $1 \leq y \leq 2$: $\frac{y+2}{4}$
- $y > 2$: 1

PDF

$$f_Y(y) = \frac{1}{4} \quad \text{for } (-2,-1) \text{ and } (1,2)$$

Point mass:

$$P(Y=0) = \frac{1}{2}$$

Q7. Stock Model Approximation

$$S_{1000} = S_0 u^k d^{1000-k}$$

Require:

$$u^k d^{1000-k} \geq 1.30$$

Take logarithm:

$$k \ln u + (1000-k) \ln d \geq \ln(1.30)$$

Solve:

$$k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d}$$

Substitute values:

$$k \geq 469.4$$

Thus:

$$k \geq 470$$

Since:

$$k \sim Binomial(1000, 0.52)$$

Required probability:

$$\sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Repair Time

$$X \sim Exponential(1/2)$$

PDF:

$$f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1} \approx 0.3679$$

(b)

$$P(X \geq 10 | X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

$$= \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y)\, dy\, dx$$

Differentiate using Leibniz rule:

$$\frac{d}{dz}\sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final PDF:

$$f_Z(z) = z \int_{-z}^{z} f_X(x) \frac{f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of M_n and M_{n+1}

$$M_n = \max(X_1, ..., X_n)$$

$$P(M_n \le x) = F(x)^n$$

Two cases:

Case 1: $x \le y$

$$P(M_n \le x, M_{n+1} \le y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \le x, M_{n+1} \le y) = F(y)^{n+1}$$

Final Joint CDF

$$P(M_n \le x, M_{n+1} \le y) = \begin{cases} F(x)^n F(y), & x \le y \\ F(y)^{n+1}, & x > y \end{cases}$$

End of Tutorial-2 Complete Step-by-Step Scribe

If you want, I can now create:

- 🔹 A 2-page ultra-fast revision sheet
- 🔹 Only formulas + key tricks
- 🔹 Practice exam-type problems for CSE 400

Since your Probability exam is close, tell me how you want to revise next.

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CSE 400: Fundamentals of Probability in Computing

Tutorial-2 – Exam Preparation Lecture Scribe

Q1. Random Point on a Line Segment

Interpretation of the Statement

A point is chosen at random on a line segment of length L.

Let:

- The point be chosen at distance x from one end,
- $0 < x < L$.

This forms two segments:

- x
- $L - x$

The ratio of the shorter to the longer segment is:

$$R = \frac{\min(x, L-x)}{\max(x, L-x)}$$

We want:

$$P(R < 1/4)$$

Logical Case Division

Divide based on midpoint $L/2$.

Case 1: $x \leq L/2$

Shorter segment = x

Longer segment = $L - x$

$$\frac{x}{L-x} < \frac{1}{4}$$

Multiply:

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > L/2$

Shorter segment = $L - x$

Longer segment = x

$$\frac{L-x}{x} < \frac{1}{4}$$

Multiply:

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\left(\frac{L}{5} - 0\right) + \left(L - \frac{4L}{5}\right) = \frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since the point is chosen uniformly over total length L:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Equal Posterior Probability (Bayes' Theorem)

Given

White region:

$$X|W \sim N(4,4)$$

So,

$$\sigma_1 = 2$$

Black region:

$$X|B \sim N(6,9)$$

So,

$$\sigma_2 = 3$$

Prior probabilities:

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observed reading:

$$X = 5$$

Condition for equal probability of error:

$$P(B|X=5) = P(W|X=5) = \frac{1}{2}$$

Apply Bayes' Theorem

$$P(B|X=5) = \frac{\alpha f(5|B)}{\alpha f(5|B) + (1-\alpha) f(5|W)} = \frac{1}{2}$$

Normal Density Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute densities:

$$f(5|B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5|W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Solve for α

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Q3. Optimal Location of Fire Station

(a) Uniform Distribution

$$X \sim Uniform(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[|X - a|]$$

Split integral:

$$\int_0^a (a-x)\frac{1}{A}dx + \int_a^A (x-a)\frac{1}{A}dx$$

Differentiate:

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a - 2A)$$

Set to zero:

$$4a = 2A$$

$$a = \frac{A}{2}$$

(b) Exponential Distribution

$$f_X(x) = \lambda e^{-\lambda x}$$

Derivative:

$$\frac{d}{da}E[|X - a|] = 1 - 2e^{-\lambda a}$$

Set equal to zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = 1/2$$

$$-\lambda a = \ln(1/2) = -\ln 2$$

$$a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

Two types of batteries.

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$P(X > t + s | X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Poisson Distribution

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

Independent.

We need:

$$P(X = k | X + Y = n)$$

Step 1

$$P(X = k, Y = n - k) = P(X = k)P(Y = n - k)$$

Substitute PMFs:

$$= \frac{e^{-\lambda_1}\lambda_1^k}{k!} \cdot \frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}$$

Denominator:

$$P(X + Y = n) = \frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1 + \lambda_2)^n}{n!}$$

Final Simplified Form

$$P(X = k | X + Y = n) = \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Q6. Transformation of Random Variable

$$X \sim Uniform[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

Piecewise definition of $g(x)$:

- x for $[-2, -1]$
- 0 for $(-1, 1)$

- x for $[1, 2]$

CDF

- $y < -2$: 0
- $-2 \leq y < -1$:

$$F_Y(y) = \frac{y+2}{4}$$

- $-1 \leq y < 0$:

$$F_Y(y) = \frac{1}{4}$$

At $y = 0$:

$$P(Y = 0) = \frac{2}{4} = \frac{1}{2}$$
$$F_Y(0) = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$$

- $0 < y < 1$: $3/4$
- $1 \leq y \leq 2$:

$$F_Y(y) = \frac{y+2}{4}$$

- $y > 2$: 1

PDF

$$f_Y(y) = \frac{1}{4} \quad \text{for } (-2, -1) \text{ and } (1, 2)$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

Q7. Stock Model

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \geq 1.30$$

Take logarithm:

$$k \ln u + (1000 - k) \ln d \geq \ln(1.30)$$

Solve:

$$k \geq \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d}$$

Using values:

$$k \geq 469.4$$

Thus:

$$k \geq 470$$

Since:

$$k \sim Binomial(1000, 0.52)$$

Required probability:

$$\sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Repair Time

$$X \sim Exponential(1/2)$$

$$f(x) = \frac{1}{2} e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1} \approx 0.3679$$

(b)

$$P(X \geq 10|X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = P(X^2 + Y^2 \leq z^2)$$

$$= \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate:

$$\frac{d}{dz}\sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final PDF:

$$f_Z(z) = z \int_{-z}^{z} f_X(x) \frac{f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of M_n and M_{n+1}

$$M_n = \max(X_1, \ldots, X_n)$$

$$P(M_n \leq x) = F(x)^n$$

Case 1: $x \leq y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leq x, M_{n+1} \leq y) = F(y)^{n+1}$$

Final Joint CDF

$$P(M_n \le x, M_{n+1} \le y) = \begin{cases} F(x)^n F(y), & x \le y \\ F(y)^{n+1}, & x > y \end{cases}$$

End of Complete Tutorial-2 Lecture Scribe

If you want next, I can prepare:

- A **compressed 2-page revision sheet**
- A **problem-type wise strategy sheet**
- Or **exam-style practice questions with full derivations**

Tell me what you need for your CSE 400 revision.